Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **ASM FinTech Developer**
b. Team leader name: **Abhinav Mukwane**
c. Problem Statement: **Intelligent Risk, Fraud & Financial Experiences (BFSI).**
   FinRiskLens — evidence-based detection of tampered loan documents and the fraud rings behind them

Brief about the idea

₹36,014 cr

involved in bank frauds in FY25

₹33,148 cr

of it in loans — the largest category

Source: RBI Annual Report 2024-25 (date of reporting)

Loans are decided on uploaded documents. A statement, salary slip or ITR can be edited in minutes, and one fabricator can supply many applicants.

FinRiskLens is an AI investigator for those documents. It reads every page with Gemini, checks the arithmetic, compares the statement with the bank's own Account Aggregator record, links applications that share a template, file or salary account, and lets a Gemini Pro agent investigate.

Every finding names the risk in rupees and links to the exact row, field or page — **the credit officer confirms or dismisses it in one click.**

The document

Balances, page totals, slip arithmetic and file history

The bank's own record

The statement against the Account Aggregator feed

The network

Shared files, templates and salary accounts form rings

AI judgement, grounded

Gemini Pro investigates; every claim cites the case's rows

Opportunities

How it differs from existing tools

- **Statement analysers** compute metrics for one application at a time — FinRiskLens returns findings, each with its evidence
- **Arithmetic checks** pass a careful forgery — FinRiskLens also compares it with the bank's own record
- **Single-case review** misses organised fraud — the network lens links applications into rings

How it solves the problem

- **Fast:** a statement is read and checked in under a minute
- **Verifiable:** every finding opens the document at its page and row
- **Decisive:** the officer confirms or dismisses each finding with a note; genuine applicants move on quickly
- **Policy-aware:** findings cite the lender's credit policy and fraud typology

USP

- **Grounded AI:** a claim that cannot point to rows of the case is set aside — and shown
- **Bank-record cross-check** through Account Aggregator
- **Fraud-ring discovery** across every application
- **Measured:** accuracy scored row by row against ground truth
- **Documents stay in India:** read and stored in Mumbai

Who gains

Credit officers

Findings with evidence, ready to act on

Fraud teams

Rings across the whole portfolio

Honest applicants

No findings — a faster decision

List of features offered by the solution

Document reading

- Gemini Flash in Mumbai reads every page in parallel
- Numbers copied as printed, never corrected
- A page that does not add up is read again with more thinking

Nine deterministic checks

- Running balance, page carry-over, stated totals
- Slip arithmetic, round-tripping, window dressing
- Pass-through, PDF file history, bank record

Bank-record cross-check

- Statement vs the bank's Account Aggregator feed
- Changed amounts, added and missing transactions
- Net effect on credits and debits in ₹

Fraud-ring discovery

- Same file, same template, same salary account
- Union-find joins linked applications
- Rings page and a finding on every member

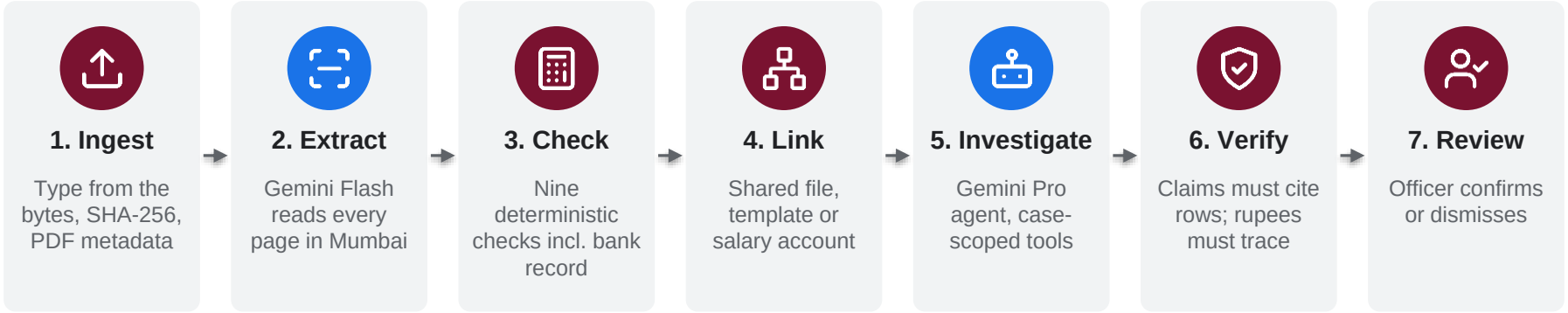
AI investigator, grounded

- Gemini Pro with tools bound to one case
- Every claim cites rows; amounts must trace
- Rejected claims shown, never hidden

Policy search and review

- Hybrid search: keyword + vector + Gemini rerank
- Findings cite the credit-policy clause
- Confirm, dismiss, reopen with notes

Process flow diagram



Credit officer

Opens a case, uploads the applicant's documents, reviews each finding with its evidence

Applicant (MSME)

Shares bank data through Account Aggregator consent; GST, ITR and Udyam are fetched to verify against

Fraud team

Works the Rings page: linked applications reviewed together before any is decided

Wireframe — anatomy of a finding

HIGH Tampering STATEMENT_VS_AA_MISMATCH Deterministic

Statement disagrees with the bank's own record

91 of 93 rows agree with the bank's Account Aggregator record, with 2 amounts changed and 4 transactions left out. Net effect: ₹3,65,800 more in credits and ₹1,30,000 less in debits than the bank recorded.

↳ **Assess the application on the Account Aggregator data and ask the applicant to explain every difference.**

↗ statement.pdf, p. 2 · row 28 — bank recorded ₹1,09,500

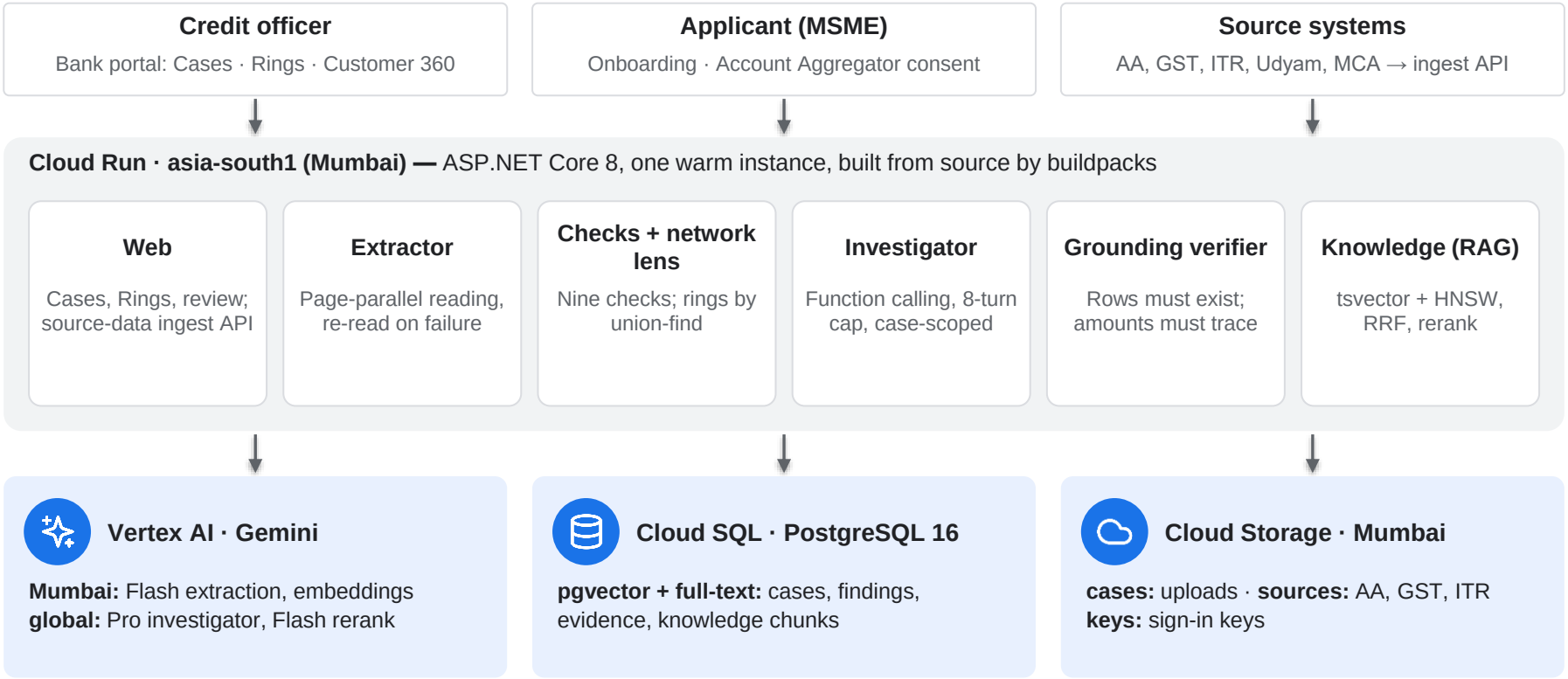
↗ statement.pdf, p. 3 · row 54 — bank recorded ₹73,400

↗ statement.pdf, p. 1 · 05 Jun NACH EMI ₹32,500 — not on the statement

Note for the case file (optional) | Confirm | Dismiss

1. **Severity, category, type and source**
   Deterministic or AI · verified
2. **What is wrong, in rupees**
   Plain language a credit officer can act on
3. **What to do next**
   Names the policy clause when one applies
4. **Evidence links**
   Open the document at that page; rows, fields or policy
5. **Review in one click**
   Confirmed and dismissed findings survive re-runs

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google Cloud

- Vertex AI — Gemini 3.5 Flash (extraction), Gemini 3.1 Pro (investigator), Gemini 3.8 Flash (rerank), gemini-embedding-001
- Cloud Run (buildpacks, no Dockerfile) · Cloud Build
- Cloud SQL for PostgreSQL 16 + pgvector · Cloud Storage

AI techniques

- Multimodal structured output: PDF and image to a fixed schema
- Function-calling agent with tools bound server-side to one case
- Grounding verifier · hybrid RAG: full-text + HNSW vectors, reciprocal rank fusion, LLM rerank

Application

- .NET 8 · ASP.NET Core MVC · Razor + Bootstrap 5
- EF Core 8 + Npgsql · Autofac · Serilog
- PdfPig (PDF metadata and page split) · background worker on Channel<T>

Data, quality and trust

- Account Aggregator (Finvu) · GST · ITR · Udyam · MCA as sources of truth
- QuestPDF synthetic cases with ground truth · 63 xUnit tests
- Secrets outside the public repo · encrypted DB connections only

Estimated implementation cost (optional)

₹ 18 L
One-time — production hardening, security and compliance audit

₹ 32 L / yr
Recurring — Google Cloud, Gemini usage, data access, support

~ ₹ 1.14 Cr
3-year total cost of ownership

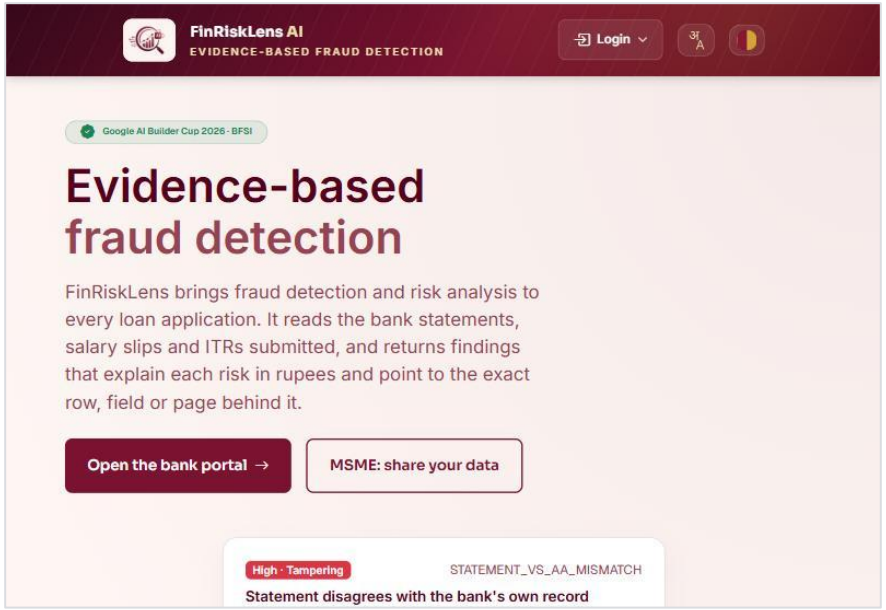
Estimate for one lender, about 1 lakh applications a year, on Google Cloud in Mumbai

Cost head	Type	Estimate (INR)
Production hardening — 3-member team, about 4 months	One-time	₹ 14 L
Security testing and RBI / DPDP compliance audit	One-time	₹ 4 L
Google Cloud — Cloud Run (HA), Cloud SQL (HA), Cloud Storage	Annual	₹ 8 L
Vertex AI — Gemini extraction, investigation and embeddings	Annual	₹ 6 L
Data access — Account Aggregator and registry APIs	Annual	₹ 10 L
Maintenance, support and model re-evaluation	Annual	₹ 8 L

Snapshots of the prototype

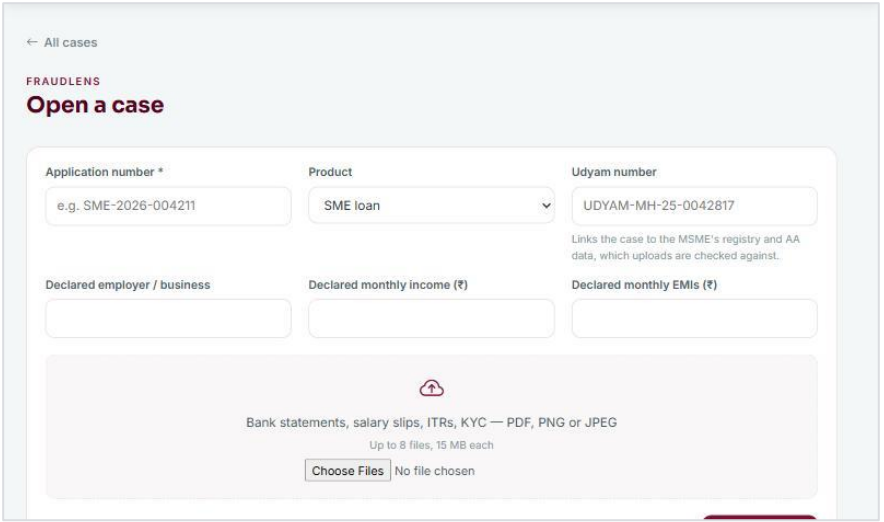
Home page

What FinRiskLens does, with a real finding



Opening a case

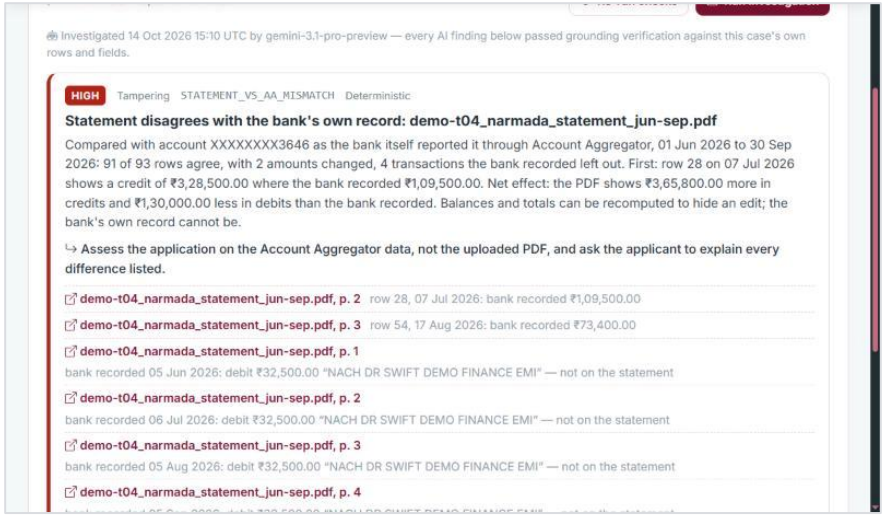
Application details, Udyam number and the documents



Snapshots of the prototype

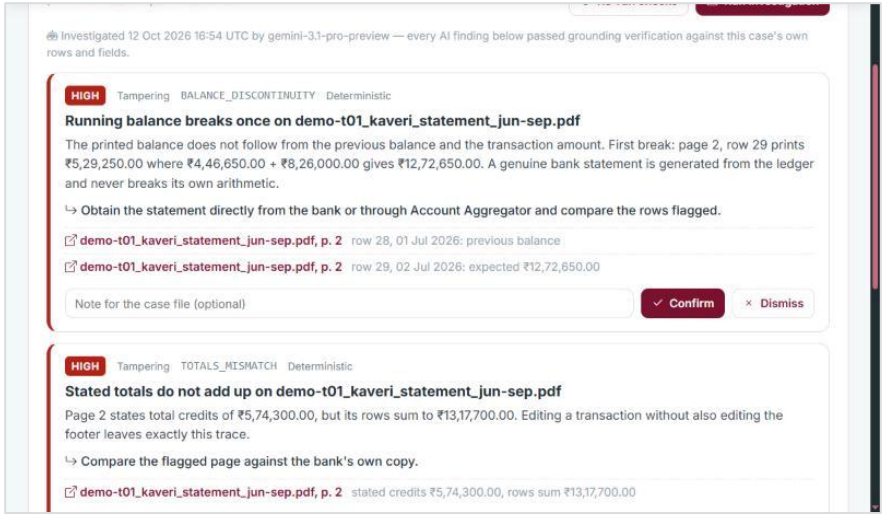
Statement vs the bank's own record

A forgery that passes every arithmetic check



The document's own arithmetic

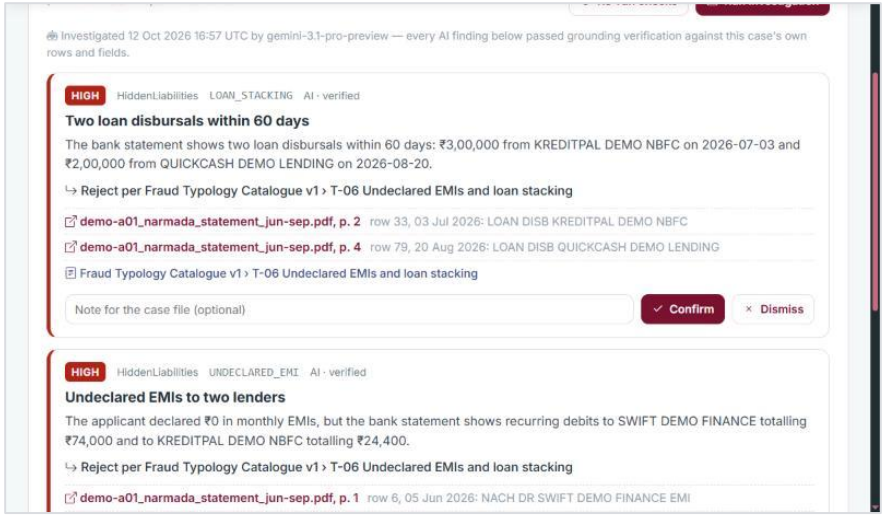
A balance break and page totals that disagree



Snapshots of the prototype

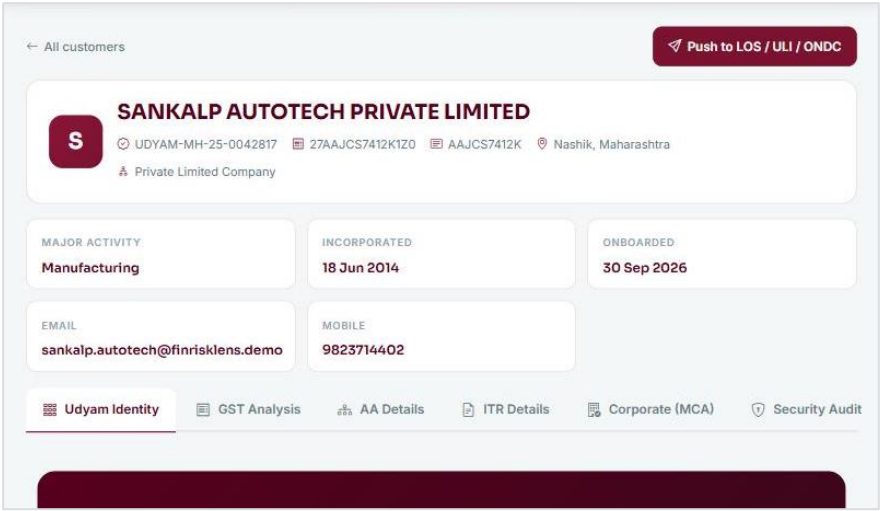
AI investigator, grounded

Undeclared EMIs and loan stacking, cited to rows and policy



Verification sources

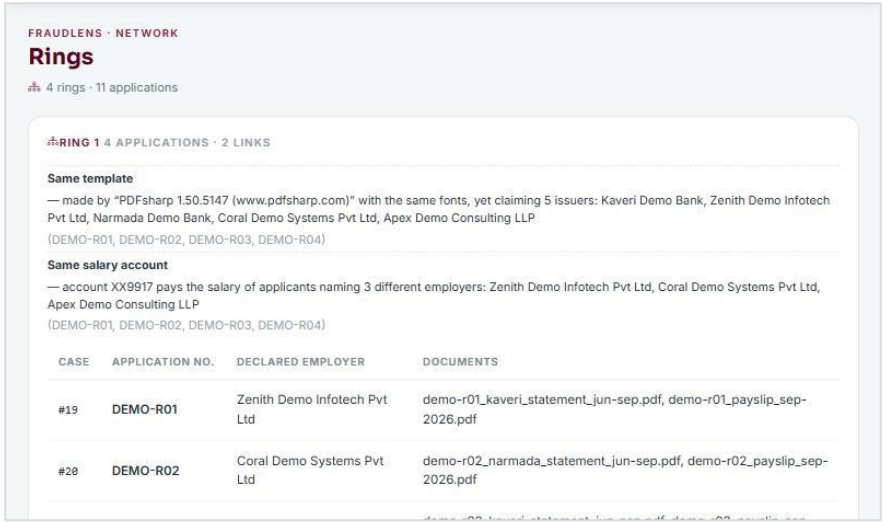
Udyam, GST, Account Aggregator, ITR and MCA per MSME



Snapshots of the prototype

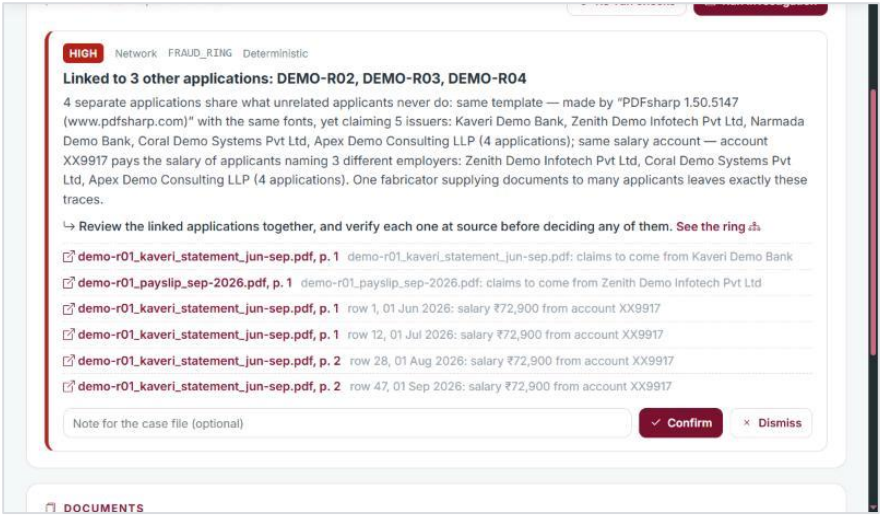
Rings

One template claiming five issuers; one salary account



The ring finding on each member

Its own documents and salary rows as evidence



Prototype Performance report/Benchmarking

99.8–100%
of 1,298 statement rows read exactly — two live runs

20 / 20
demo cases raise exactly their expected findings

0
false alarms on genuine applicants — checks and AI

5 / 5
expected AI findings found by the investigator

How it is measured: synthetic documents with known ground truth, read by live Gemini, scored row by row, checks run on Gemini's own reading

Check	Manual review	FinRiskLens
Statement arithmetic	Spot-checked by eye	Every row and page: balance, carry-over, totals
Statement vs bank	Ask for a bank-certified copy	Automatic when Account Aggregator is linked
Edited files	Rarely examined	Producer, creator and edit dates read on upload
Links across applications	One application at a time	Shared file, template or salary account → rings
AI reliability	—	Every claim cites rows; rejected claims shown
Speed	Analyst time per file	10–60 s per statement · 12–43 s per investigation

Additional Details/Future Development (if any)

Registry cross-checks

- Uploaded ITR against the filed return
- Claimed employer against the MCA register
- Edge cases: password-protected and scanned statements

Evidence highlighting

- Highlight the exact row on the PDF page
- Bounding boxes already in the extraction contract
- Side-by-side PDF and bank record view

Case memory and policy Q&A

- Confirmed findings feed the knowledge base
- Policy questions answered with citations
- Refusal when the policy is silent

Scale and integration

- Entity link tables for portfolio-scale rings
- Findings pushed to the LOS with the loan case
- Per-lender policies and thresholds

Provide links to your:

- GitHub Public Repository: https://github.com/abhinavmukwane/FinRiskLensAI-RAG-V2
- Demo Video Link (3 Minutes): **add the link after uploading the video**
- Final Product Link: https://finrisklensai.cloud.run

For the evaluation panel

Bank portal: https://finrisklensai.cloud.run/Auth/BankLogin — credentials are shared with the panel directly.

Every case on the live site is synthetic: fictional banks, employers and applicants, each page stamped SYNTHETIC.

In the repository

README — how it works and every measurement · Doc/ARCHITECTURE.md · Doc/10_FRAUD_LENS_BUILD.md — build plan, deploy command

Run it yourself: dotnet test (63 tests) · DemoData eval (live extraction scored against ground truth)

Demo cases on the live site

Case	What it is	What FinRiskLens finds
DEMO-G01 … G05	Genuine applicants; G05 has Account Aggregator linked	Nothing — clean applicants move on
DEMO-T01	A receipt inflated tenfold in Word	Balance break, totals mismatch, file edited in Word
DEMO-T02	Page 3 swapped in from a healthier quarter	Page carry-over and balance breaks, merged in iLovePDF
DEMO-T03	Salary slip with net pay typed higher, made in Canva	Slip arithmetic; AI: slip vs salary actually credited
DEMO-T04	Careful forgery: receipts tripled, EMIs deleted, totals recomputed	Statement vs the bank's Account Aggregator record
DEMO-B01 … B03	Round-tripping · window dressing · pass-through	Each pattern, with the rows behind it
DEMO-A01	Monthly EMIs to two lenders never declared	AI: undeclared EMIs, loan stacking — cited to policy
DEMO-A02	Weekly betting, monthly crypto, a bounced EMI	AI: gambling and crypto exposure, the bounce
DEMO-R01 … R04	Four applicants from one fabricator	A ring: one template, one salary account
DEMO-R05, R06	One statement file in two applications	A ring: the same file, byte for byte

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Thank you

